EXERCISE 16

Pitchfork Phase Line and Stability Regimes

Original statement · ProblemsCh2.pdf, p. 4

Phase line • Stability • Explicit solution

PROBLEM STATEMENT

Consider the autonomous equation

$$\frac{dx}{dt} = x\left(\kappa^2 - x^2\right)$$

with a general initial condition $x(0) = x_0$.

(i) Calculate the stationary solutions, which satisfy $dx/dt = 0$;
(ii) Calculate and sketch the solutions for $t > 0$, discussing the different behaviors which can be obtained in terms of the relative values of $\kappa > 0$ and x_0, with the help of the uniqueness theorem.

Source: ProblemsCh2.pdf, Exercise 16 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Autonomous equation $\dot{x} = f(x)$ with $f(x) = x(\kappa^2 - x^2) = \kappa^2 x - x^3$ and parameter $\kappa > 0$. We have

$$f'(x) = \kappa^2 - 3x^2.$$

Both f and f' are polynomials, hence continuous on $\mathbb{R}$: the IVP has a unique solution for every $x_0 \in \mathbb{R}$, on a maximal interval of existence. The right-hand side is odd, $f(-x) = -f(x)$, so if $x(t)$ is a solution then $-x(t)$ is also a solution (symmetry $x \to -x$).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why these methods: (a) the phase line (sign of f between equilibria) and the no-crossing consequence of uniqueness give the qualitative picture for every x_0 without solving; (b) the equation is a Bernoulli equation ($\alpha = 3$, see Problem 2.8), so the substitution $w = x^{-2}$ turns it into a linear equation and gives the exact solution to confirm the qualitative picture and to find timescales.

Change of variables written explicitly: $w = x^{1-\alpha} = x^{-2}$ (valid for $x \neq 0$), $w' = -2x^{-3}x'$, $w(0) = 1/x_0^2$.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The test of the notes: a stationary point x^* with $f'(x^*) > 0$ is unstable, with $f'(x^*) < 0$ stable. Here $f(x) = \kappa^2 x - x^3$ and $f'(x) = \kappa^2 - 3x^2$: $f'(0) = \kappa^2 > 0$ (unstable) and $f'(\pm\kappa) = \kappa^2 - 3\kappa^2 = -2\kappa^2 < 0$ (stable). Between consecutive zeros the sign of f fixes the direction of motion, as in the sketch of the notes, and Picard's theorem (f and f' continuous everywhere) makes the stationary solutions barriers that no other solution crosses.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): stationary solutions

Solve $f(x^*) = x^*(\kappa - x^*)(\kappa + x^*) = 0$ using $\kappa^2 - x^2 = (\kappa - x)(\kappa + x)$:

$$x^* = 0, \qquad x^* = \kappa, \qquad x^* = -\kappa.$$

Stability from $f'(x^*)$: $f'(0) = \kappa^2 > 0$ (unstable); $f'(\pm\kappa) = \kappa^2 - 3\kappa^2 = -2\kappa^2 < 0$ (asymptotically stable). Therefore the constant solutions are $x \equiv 0$ (unstable) and $x \equiv \pm\kappa$ (stable).

Part (ii), step 1: phase line from uniqueness

By uniqueness solutions cannot cross the equilibria, so the equilibria $-\kappa, 0, \kappa$ split $\mathbb{R}$ into four invariant intervals. Sign of $f = x(\kappa - x)(\kappa + x)$:

Region	**Signs of $x, \kappa - x, \kappa + x$**	**Sign of f**	**Motion**
$x > \kappa$	$+, -, +$	$-$	decreases to κ
$0 < x < \kappa$	$+, +, +$	$+$	increases to κ
$-\kappa < x < 0$	$-, +, +$	$-$	decreases to $-\kappa$
$x < -\kappa$	$-, +, -$	$+$	increases to $-\kappa$

In each region the solution is monotone and bounded by the neighboring equilibria, hence global in time, with a limit that is an equilibrium: all solutions with $x_0 > 0$ tend to κ, all with $x_0 < 0$ tend to $-\kappa$, and $x_0 = 0$ stays at 0.

Part (ii), step 2: exact solution

Let $x_0 \neq 0$ (then $x(t) \neq 0$ for all t, by uniqueness). Compute $w = x^{-2}$ using the chain rule:

$$w' = -2x^{-3}x' = -2x^{-3}\left(\kappa^2 x - x^3\right) = -2\kappa^2 x^{-2} + 2 = -2\kappa^2 w + 2.$$

Linear equation $w' + 2\kappa^2 w = 2$. Integrating factor: $\int 2\kappa^2\,dt = 2\kappa^2 t$, $\mu = e^{2\kappa^2 t}$, $\mu(0) = 1$, and $(\mu w)' = 2e^{2\kappa^2 t}$. Barrow from 0 to t:

$$w(t)\,e^{2\kappa^2 t} - w(0) = \int_0^t 2e^{2\kappa^2 s}\,ds = \frac{1}{\kappa^2}\left[e^{2\kappa^2 s}\right]_0^t = \frac{1}{\kappa^2}\left(e^{2\kappa^2 t} - 1\right).$$

Hence, with $w(0) = 1/x_0^2$,

$$w(t) = \frac{1}{\kappa^2} + \left(\frac{1}{x_0^2} - \frac{1}{\kappa^2}\right)e^{-2\kappa^2 t} = \frac{x_0^2 + (\kappa^2 - x_0^2)\,e^{-2\kappa^2 t}}{\kappa^2 x_0^2}.$$

Since $x = \pm w^{-1/2}$ with the sign fixed by x_0 (a solution cannot change sign), and $x(0) = x_0$:

$$x(t) = \frac{\kappa\, x_0}{\sqrt{x_0^2 + (\kappa^2 - x_0^2)\,e^{-2\kappa^2 t}}}, \qquad x_0 \neq 0.$$

(For $x_0 = 0$ the solution is $x \equiv 0$, which the formula does not cover.) Check: at $t = 0$, the square root is $\sqrt{\kappa^2} = \kappa$, so $x(0) = x_0$.

Part (ii), step 3: behavior for $t > 0$

Write $E(t) = e^{-2\kappa^2 t} \in (0, 1]$ for $t \geq 0$. The radicand is $x_0^2(1 - E) + \kappa^2 E$, a convex combination of $x_0^2 > 0$ and $\kappa^2 > 0$, hence positive for all $t \geq 0$: the solution exists for all $t > 0$. As $t \to \infty$, $E \to 0$, the radicand $\to x_0^2$ and

$$x(t) \to \frac{\kappa\, x_0}{|x_0|} = \kappa\ \mathrm{sgn}(x_0).$$

Initial condition	**Behavior for $t > 0$**	**Limit as $t \to \infty$**
$x_0 > \kappa$	decreasing, stays above κ	κ
$x_0 = \kappa$	constant $x \equiv \kappa$	κ
$0 < x_0 < \kappa$	increasing, stays below κ	κ

Initial condition	**Behavior for $t > 0$**	**Limit as $t \to \infty$**
$x_0 = 0$	constant $x \equiv 0$	0 (unstable)
$-\kappa < x_0 < 0$	decreasing, stays above $-\kappa$	$-\kappa$
$x_0 = -\kappa$	constant $x \equiv -\kappa$	$-\kappa$
$x_0 < -\kappa$	increasing, stays below $-\kappa$	$-\kappa$

Shape of the sketch: for $0 < x_0 < \kappa$ the curve is sigmoidal if $x_0 < \kappa/\sqrt{3}$ (inflection point where $x'' = f'(x)\,x' = 0$, i.e. $f'(x) = \kappa^2 - 3x^2 = 0$, at $x = \kappa/\sqrt{3}$, where the slope is maximal), and concave without inflection if $x_0 \geq \kappa/\sqrt{3}$. The curves for $x_0 < 0$ are mirror images ($x \to -x$).

Behavior backward in time (for completeness): the radicand vanishes when $E = x_0^2/(x_0^2 - \kappa^2)$, which requires $|x_0| > \kappa$ and happens at $t = -\frac{1}{2\kappa^2} \ln \frac{x_0^2}{x_0^2 - \kappa^2} < 0$. So solutions with $|x_0| > \kappa$ blow up in finite negative time, while those with $0 < |x_0| < \kappa$ exist for all $t < 0$ and tend to 0 as $t \to -\infty$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the ODE by differentiating the explicit solution. With $D(t) = x_0^2 + (\kappa^2 - x_0^2)e^{-2\kappa^2 t}$: $x = \kappa x_0 D^{-1/2}$, $x' = -\frac{1}{2}\kappa x_0 D^{-3/2} D'$, and $D' = -2\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}$. Also $\kappa^2 - x^2 = \kappa^2 - \frac{\kappa^2 x_0^2}{D} = \frac{\kappa^2(D - x_0^2)}{D} = \frac{\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}}{D}$, so $x(\kappa^2 - x^2) = \kappa x_0 D^{-1/2} \cdot \frac{\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}}{D}$. This equals x': $-\frac{1}{2}\kappa x_0 D^{-3/2} \cdot \left(-2\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}\right) = \kappa x_0 D^{-3/2}\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}$. They agree.

Numerical check (computed): for $\kappa = 1$, a fourth-order Runge-Kutta integration over $t \in [0, 5]$ (20000 steps) matches the closed form to better than 10^{-14} for $x_0 = 0.3,\ 2,\ -0.5,\ -3$. At $t = 5$ the distance to $\kappa = 1$ is 2.3×10^{-4} for $x_0 = 0.3$ and 1.7×10^{-5} for $x_0 = 2$, consistent with the time constant $1/(2\kappa^2) = 0.5$.

Time scale and dimensions: the linearization at $x = \pm\kappa$ has rate $f'(\pm\kappa) = -2\kappa^2$, so the approach to $\pm\kappa$ is exponential with time constant $1/(2\kappa^2)$ (visible in the factor $e^{-2\kappa^2 t}$), and at $x = 0$ the repulsion rate is κ^2. The equation is written in non-dimensional form: κ^2 is a rate with units $[t]^{-1}$, and x is measured in units for which the cubic coefficient equals 1.

Bifurcation context (Robinson, Section 7.6): for the parameter $\mu = \kappa^2$ the equation $\dot{x} = \mu x - x^3$ has one equilibrium $x = 0$ (stable) for $\mu < 0$, and for $\mu > 0$ the origin becomes unstable and two stable branches $\pm\sqrt{\mu}$ appear: a supercritical pitchfork. Here $\kappa > 0$ is the case $\mu > 0$.

RESULT

Equilibria 0 (unstable) and $\pm\kappa$ (stable). For $x_0 \neq 0$: $x(t) = \kappa x_0 / \sqrt{x_0^2 + (\kappa^2 - x_0^2)e^{-2\kappa^2 t}} \to \kappa\ \mathrm{sgn}(x_0)$ as $t \to \infty$.